Seong Youn Kim

Ph.D. student in Mathematics, Seoul National University s264575@snu.ac.kr

RESEARCH INTERESTS

Character varieties and skein algebras; mapping class groups; Hitchin representations and higher Teichmüller theory; geometric structures on manifolds; compactifications of moduli spaces; dynamics and arithmetic points. Additional interests include Diophantine equations, Coxeter groups, geometric group theory, and buildings.

EDUCATION

Seoul National University 2021–present

Ph.D. in Mathematics, in progress. Advisors: Jun Ho Whang and Gye-Seon Lee.

Seoul National University 2017–2021

B.S. in Mathematics, *summa cum laude*.

PAPERS AND PREPRINTS

1. With Sunghwan Ko, *Infinitude of integral Hitchin representations of Coxeter polygons*. Preprint, 2026. arXiv:2611.11930.

2. *Compactification of* SL(3, $\mathbb{C}$)*-Character Varieties of Surfaces via Skein Algebras*. Preprint, 2026. arXiv:2608.30217.

3. *Algebraic Mapping Class Group Rigidity*. Preprint, 2025; revised 2026. arXiv:2508.09421.

Work in progress

- *Autoequivalence of the Cobordism Categories*.
- *Scissors Congruence and the Class Number*.

SELECTED TALKS

Real Spectrum Compactification of Character Varieties and its Applications on Tits–Vinberg Space August 2025
Korea Institute for Advanced Study (KIAS).

Algebraic Mapping Class Group Rigidity January 2023; July 2025
Exchange Seminar, Seoul National University.

Survey on Mordell's Equation: SL(3)*-Character Variety over a Pants, Fock–Goncharov Conjecture and Homogeneous Dynamics* April 2024
Exchange Seminar, Seoul National University.

SELECTED CONFERENCES

One-day Festival of Group Actions, Geometry and Dynamics, KIAS, Seoul	August 2025
Geometry and Dynamics in Higher Rank Lie Groups, SLMath, Berkeley	July 2025
Low-dimensional Phenomena: Geometry and Dynamics, IHP, Paris	June 2025
Teichmüller Theory and Beyond, Seoul National University	February 2025
International Workshop on Higher Teichmüller Theory, CIM, Nankai University, Tianjin	May 2024
Hyperbolic Geometry of Numbers, KIAS, Seoul	December 2023

TEACHING

Practice Sessions for the Mathematics Qualifying Examination, SNU	Fall 2024–present
Algebra III (graduate), SNU	Spring 2025
Algebra II (graduate), SNU	Fall 2024
Algebra I (graduate), SNU	Spring 2024
Modern Algebra II, SNU	Fall 2023
Introduction to Analysis I, SNU	Spring 2023
Calculus I, SNU	Winter 2022
Calculus II, SNU	Fall 2022
Calculus II, SNU	Summer 2022
Calculus I, SNU	Spring 2022
Calculus II, SNU	Fall 2021

HONORS AND AWARDS

Outstanding Teaching Assistant, Algebra II, Seoul National University	Fall 2024
Scholarship for the Next Generation of Basic Science, Seoul National University	2023–2024
BK Research Fellowship, Seoul National University	2022
Graduate Student Instructor Scholarship, Seoul National University	2022
Outstanding Teaching Assistant, Calculus I, Seoul National University	Spring 2022
Scholarship for the Next Generation of Basic Science, Seoul National University	Fall 2021